



STRATEGY SPRINT

Reuse of Rigid Plastic

The Profit Prophets



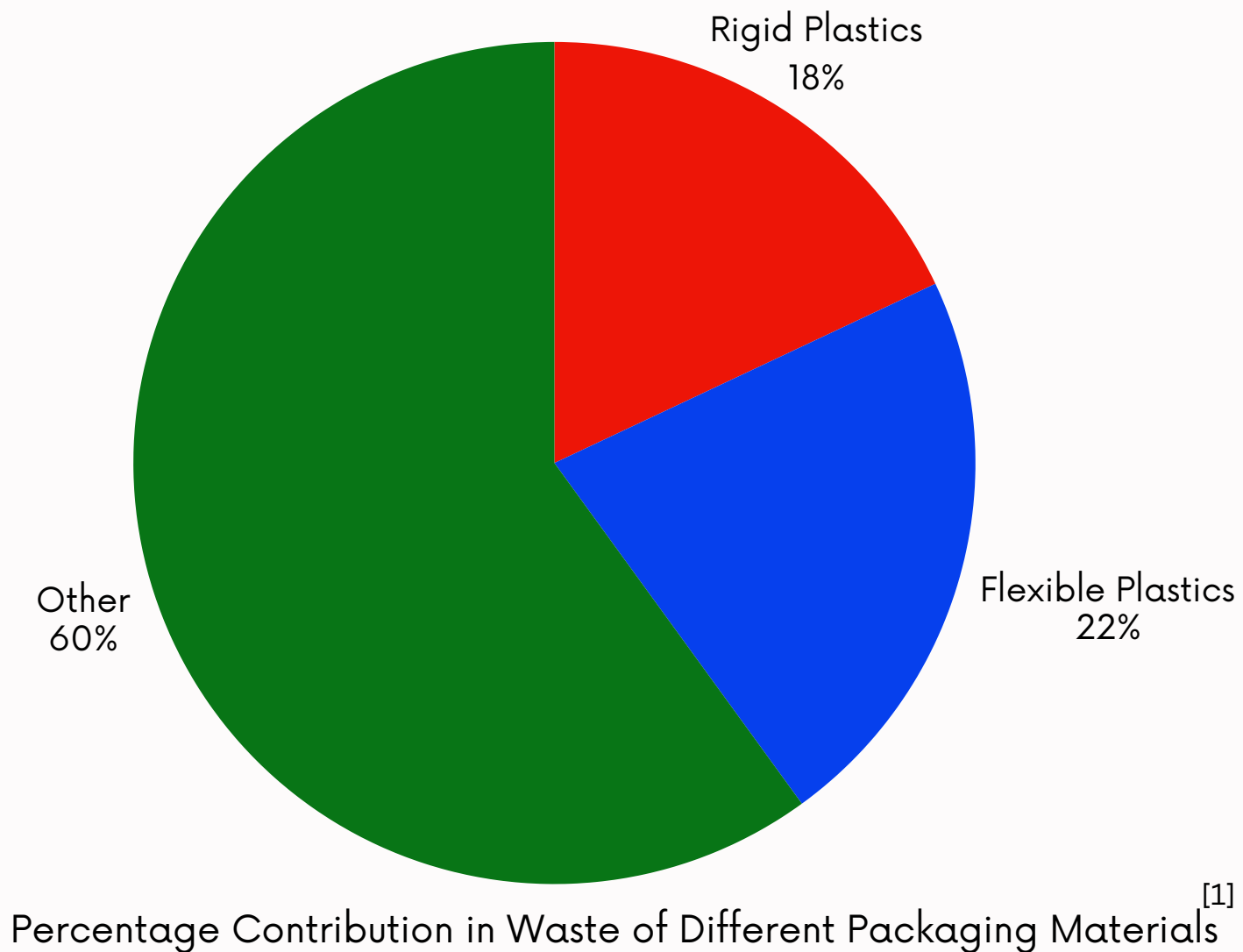
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Introduction

Globally about 400^[2] million tons of plastic waste is produced every year. Of this approximately 36^[2] percent comes from the packaging industry. Targeting this worldwide problem, our reuse-based packaging model offers a simple yet powerful solution: instead of throwing away bottles, they are collected, cleaned, refilled, and reused, reducing waste while benefiting both businesses and consumers



Industry Analysis

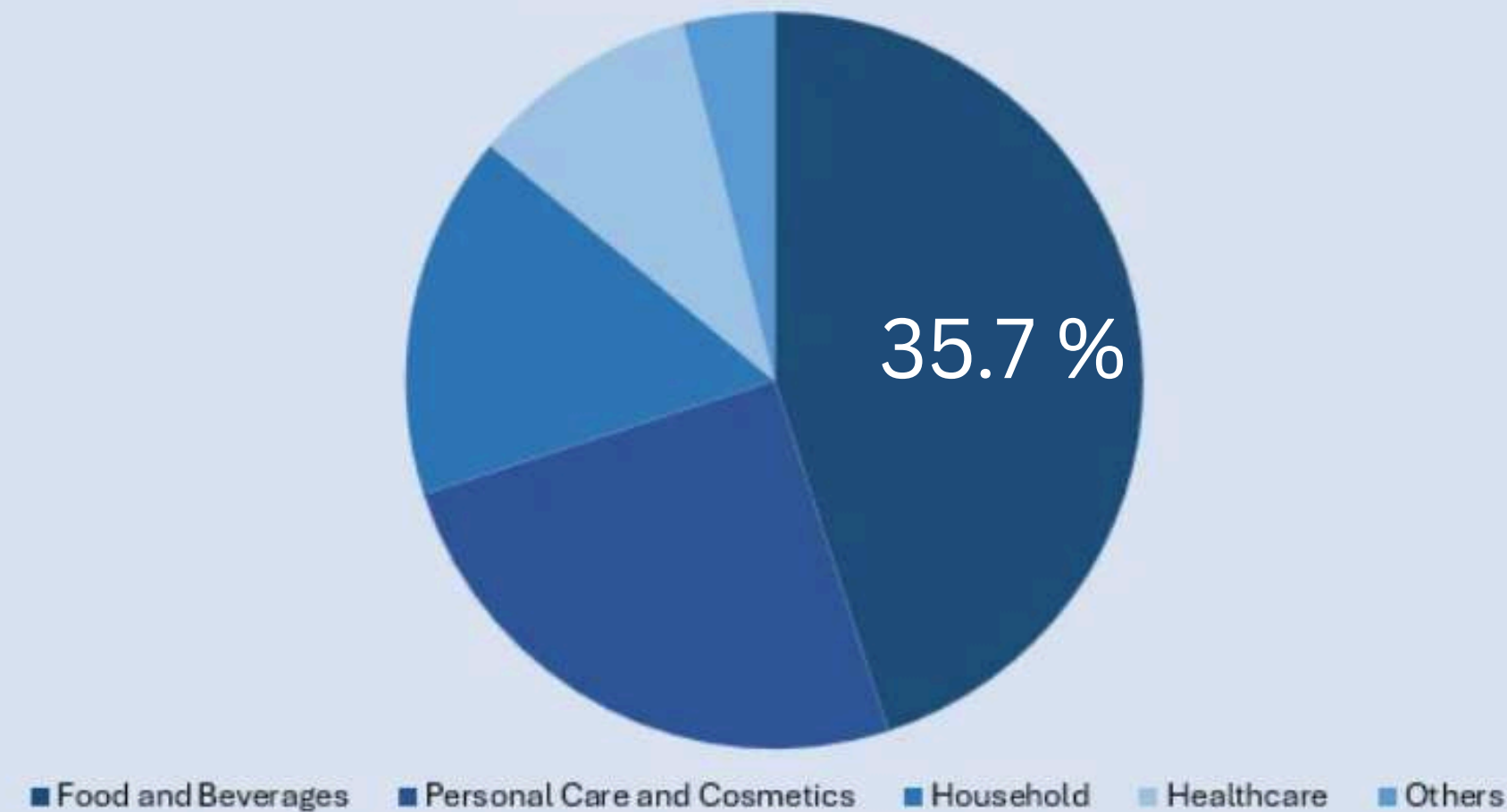
Analyzing market: Food and beverage industry at 35.7%^[1] is the largest contributor to plastic packaging waste generated across the globe.

Feasibility: Since these companies already have a distributor network in place, utilizing the same model for reverse logistics is rather not complex.

Consumer Acceptance: Coca-Cola and PepsiCo have been operating via a reuse model in various countries.

Rigid Plastic Packaging Market ^[1]

Market Share by End Use Industry, 2024 (in %)



Nestlé generates almost two million metric tons of plastic packaging waste per year. In Mexico, an estimated 75 thousand metric tons of mismanaged plastic waste is created by Coca-Cola. ^[2]

Current Delivery Mechanism

Points	Challenges	Opportunities
Retail	Consumer Behavior: Customers are used to buy-and-dispose habits Collection Infrastructure: Retailers may not have space to collect used packaging. Refilling Limitations: In-store refilling may not work for sealed products like dairy, beverages.	Return-to-Retail Model: Consumers can drop off used packaging at stores in exchange for discounts or loyalty points. Retailer Incentives: Retailers can earn incentives for collecting packaging Refill Stations for Select Products: Possible for cooking oils, milk and beverages in urban supermarkets.
E-Commerce	Consumer Behavior: Many consumers may not bother returning packaging. Cross-Contamination Risks: Used packaging may not be stored properly, affecting quality.	Deposit Refund Model: Charge a small deposit for packaging that is refunded when the customer returns it. Pickup with Next Delivery: Collect used packaging during the next delivery, reducing logistics costs. Sustainable Packaging Subscriptions: Offer returnable packaging options for frequent buyers
Quick Commerce	The fast-paced nature of quick commerce emphasizes on speed and convenience, which is conflicting with the reuse business model.	Partner with quick commerce delivery agents to collect used packaging at the time of delivery.
Direct-to-Consumer	Individual companies may lack the infrastructure to manage packaging returns effectively on their own, making scalability an issue.	Direct-to-consumer brands have more control over their processes and could develop innovative subscription-based systems to aggregate used packaging.



Our Business Model

Our Business Model



Collection

01



Retail

- Collection from small shops and local stores.
- Includes expired, unsold, or discarded items.
- Direct pick-up or scheduled collection.

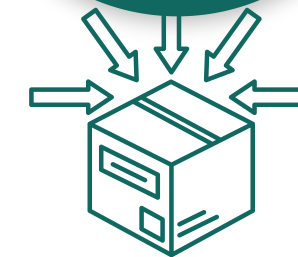
02



Supermarket kiosks

- In-store drop-off points for easy collection.
- Collected from customers or store excess.
- Periodic bulk pick-ups for efficiency.

03



Scrap Dealers and Aggregators

- Bulk sourcing from scrap traders.
- Pre-sorted materials for easier processing.
- Facilitates large-scale collection.

04

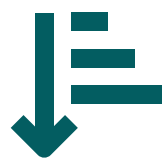


Housing Societies

- Direct collection from residential communities.
- Encourages organized waste segregation.
- Scheduled pickups for convenience.

Cleaning and Quality Check

01



Sorting

Sorting and choosing satisfactory packaging that is not damaged or broken.

02



Decontamination

Cleaning, decontaminating and sanitizing chosen packaging.

03



Quality Testing

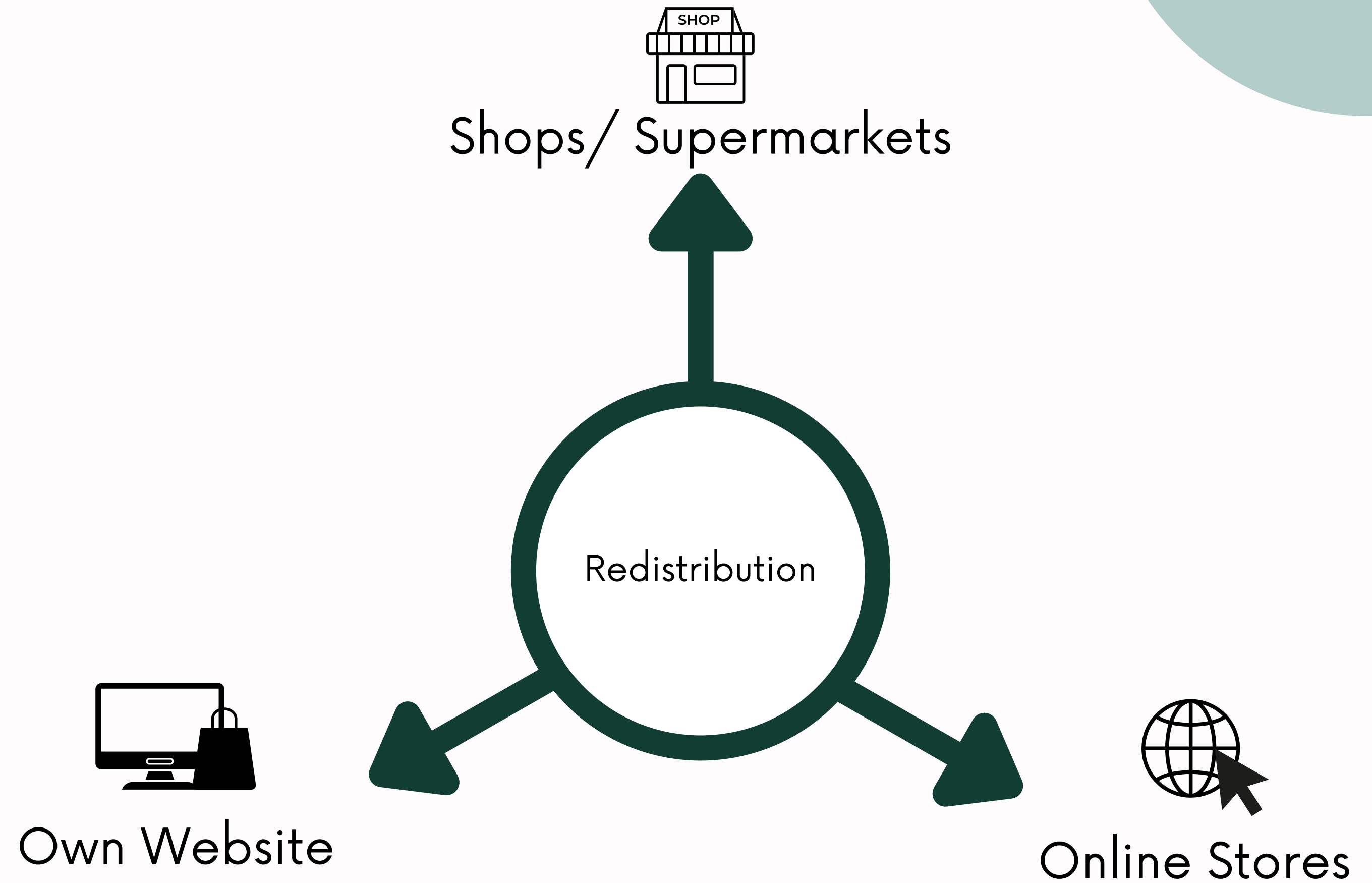
Testing cleaned packaging to ensure health and quality standards.

04



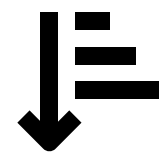
Refilling

Refilling the chosen packaging and sending it off for redistribution.



Traceability Mechanism

01



Segregating

Products post pick up can be categorized as per their nature.

02



Identification

Unique Digital identifier is provided to each shipment. Like QR code or RFID Tags or IoT Sensors

03



Updation

Updation of iimestamp, location, and ownership changes.

Challenges

- ❌ Consumer acceptance, cultural issues and awareness
- ❌ Collection and reverse logistics
- ❌ Customer stickiness to single use plastics
- ❌ Negative brand perception
- ❌ Health and safety standards compliance



It always seems impossible until it's done.
Nelson Mandela

Commercial Feasibility



Scale

Estimating viability of our model by calculating contribution margins, LTV by CAC ratios, breakeven volumes and payback period.



Investments

Capital is needed to invest in infrastructure for reuse, technology for tracking and for reverse logistics.



Incentives

Encourage customers using discounts, loyalty programs, deposit return schemes and cahsbacks.

Scale and Viability

We can use unit economics metrics to determine the scale at which the model would become viable.

Components	Reference Formula
Contribution Margin	$CM > 0$ for the model to be eventually viable ($CM = \text{Selling Price} - \text{Variable Cost}$)
CAC v/s LTV	$LTV/CAC \geq 3$ for a sustainable business
Break-even Volume	Lower the BEV faster the business reaches profitability ($BEV = \text{Fixed Cost} / \text{Contribution Margin}$)
Payback Period	Shorter Payback Period = Better Cash Flow (Time taken to recover CAC)



Investments

Durable Plastic Containers

Tamper-proof, resealable, and easy to clean.

Reverse Logistics

Efficiently collecting used packaging

Tracking Technology

QR codes, RFID chips for monitoring returns.

Washing & Sterilization Equipment

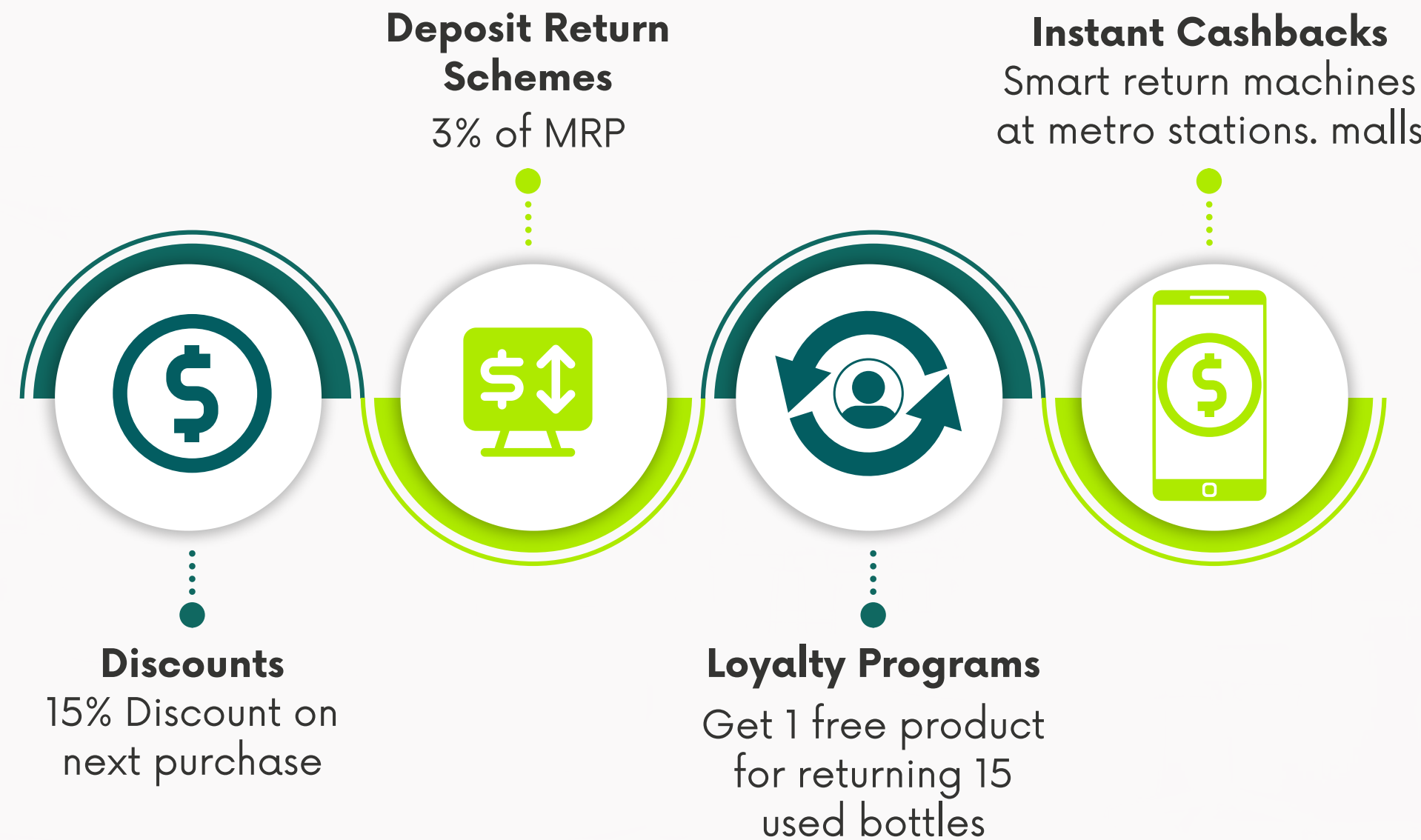
To satisfy FSSAI and FDA regulations for food grade packaging

Refill Machines

Automated refill stations for customers

Incentives

Providing incentives to consumers would encourage them to return used packaging and shift from disposable culture to a reuse mindset.



TRIPLE BOTTOM LINE CONSIDERATIONS

ECONOMIC

PROS:

- Long term savings for the company. Each bottle would cost x times more than the single use bottle but it can be used y times more than a single use.
- Higher engagement from eco-conscious consumers, especially Gen Z & Millennials.

CONS:

- High initial capital investment in cleaning, logistics, and technology.
- Supply chain complexities.
- Retailer cooperation required.

SOCIAL

PROS:

- Encourages consumers to shift from disposable culture to a reuse mindset.

CONS:

- Customers may hesitate due to habitual preference for disposables.

ENVIRONMENTAL

PROS:

- Reducing single use plastic and overall plastic consumption.
- Reduced carbon footprint by less new bottle production and less incineration of used bottles.

CONS:

- Water & energy used in cleaning & sterilization must be managed sustainably.
- Transport emissions from collection & redistribution would increase and must be optimized.

Thank You!

Our Team



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